

Exam Eligibility Checker

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COURSE: DIGITAL DESIGN FUNDAMENTALS
(EEE1200001)

Team Members and Roles

OMAR KARIMOV – Python Developer + Circuit Verse
Designer

ABDUAZIM DJAMILOV – Logic Designer

JAFARBEK NAZAROV and SOBIROV SUXROB –
Documentation / Presentation

PROBLEM

Our team identified that students often do not know if they are eligible to take the final exam.

There are multiple requirements such as attendance, assignments, fees, and academic warnings. Checking eligibility manually is confusing and time-consuming.

Attendance (A)	Assignments (B)	Fees (C)	Warning (D)	Conditionally Eligible	Not Eligible
1	All			Eligible	
2	Missing			Conditionally Eligible	
3	Missing assignments			Fees	
4	Missing Attendance			Not	Eligible
5	Missing Warning			Conditionally Eligible	
6	All condition			Not Eligible	

“ SYSTEM OVERVIEW

. Inputs:

- . Attendance met (Yes/No)
- . Assignments submitted (Yes/No)
- . Fees cleared (Yes/No)
- . Academic warning

(1=Yes,0=No) Outputs:

- . Eligible
- . Conditionally Eligible
- . Not Eligible



LOGIC DESIGN

Decision Rules:

- Eligible → All 4 conditions satisfied
- Conditionally Eligible →
 - $A' \cdot B \cdot C \cdot D'$
 - $A \cdot B' \cdot C \cdot D'$
 - $A \cdot B \cdot C' \cdot D'$
 - $A \cdot B \cdot C \cdot D$
- Not Eligible → 2 or more conditions missing

Eligible = $A \cdot B \cdot C \cdot D'$

$D = 1 \rightarrow$ warning present



TRUTH TABLE

Summary:

- 1 case → Eligible
- 4 cases → Conditionally Eligible
- 11 cases → Not Eligible

All 16 input combinations were analyzed.

Truth Table – Exam Eligibility Checker

EEE120 Digital Design Fundamentals – Project 10

1 = Yes / Met 0 = No / Not Met

A = Attendance Met

B = Assignments Submitted

C = Fees Cleared

D = Academic Warning (1 = Has Warning, 0 = No Warning)

#	A (Attendance)	B (Assignments)	C (Fees)	D (Warning)	Result
1	1	1	1	0	Eligible
2	1	1	1	1	Conditionally Eligible
3	1	1	0	0	Conditionally Eligible
4	1	1	0	1	Not Eligible
5	1	0	1	0	Conditionally Eligible
6	1	0	1	1	Not Eligible
7	1	0	0	0	Not Eligible
8	1	0	0	1	Not Eligible
9	0	1	1	0	Conditionally Eligible
10	0	1	1	1	Not Eligible
11	0	1	0	0	Not Eligible
12	0	1	0	1	Not Eligible
13	0	0	1	0	Not Eligible
14	0	0	1	1	Not Eligible
15	0	0	0	0	Not Eligible
16	0	0	0	1	Not Eligible

Summary:

- **Eligible:** 1 case
- **Conditionally Eligible:** 4 cases
- **Not Eligible:** 11 cases

Logic:

Eligible = $A \cdot B \cdot C \cdot D'$

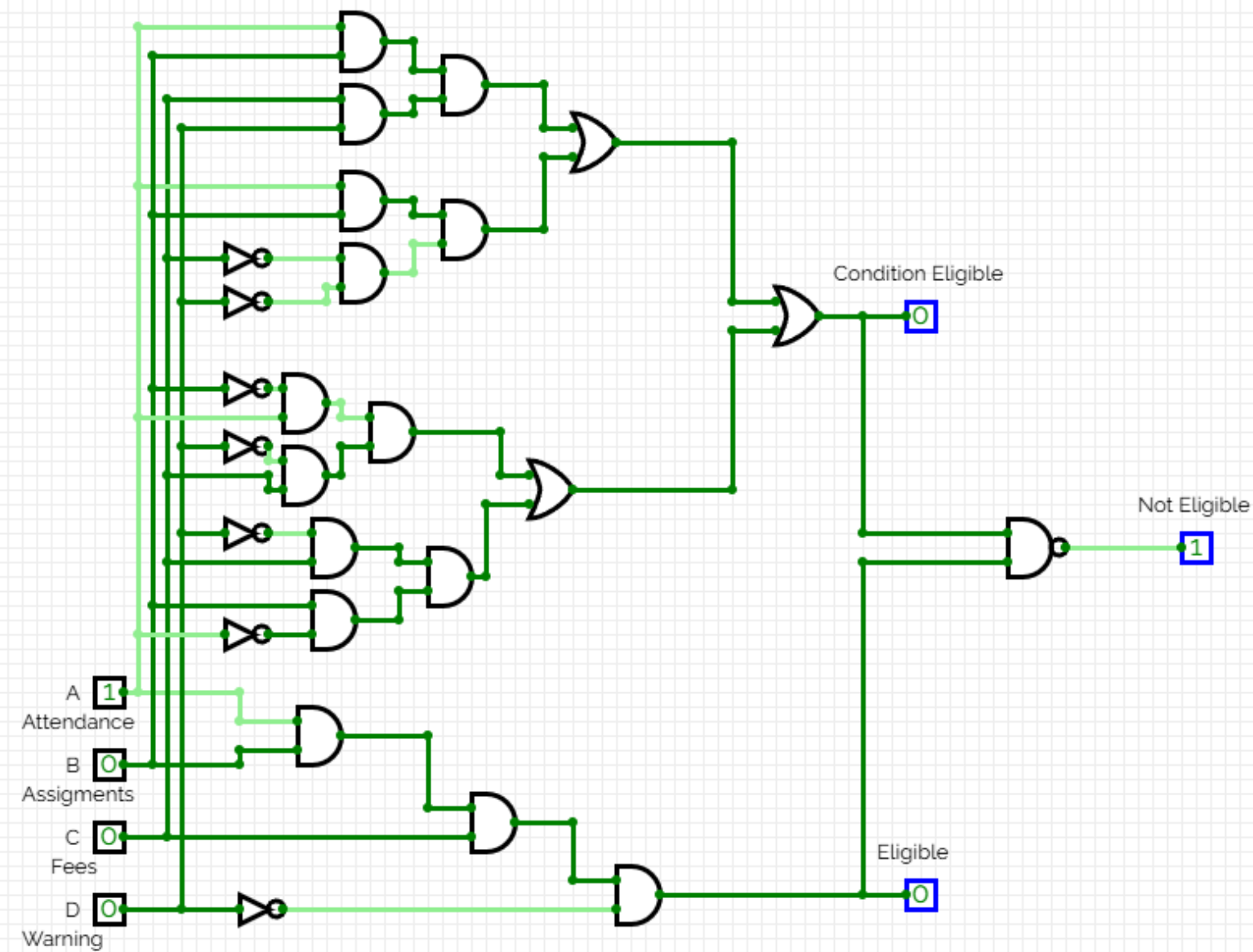
Conditionally Eligible = exactly one of A, B, C, D is 0

Not Eligible = two or more inputs are 0 or D = 1 (warning present with other issues)

Note: D = 1 means student has an academic warning.

CIRCUITVERSE DESIGN

- Built using basic logic gates
- Proper labels and clear layout
- Fully working simulation



[CircuitVerse Link](#)



PYTHON SOFTWARE PROTOTYPE

The Python code implements the same Boolean logic as the digital circuit

A = Attendance

B = Assignments

C = Fees

D = Warning

Eligible = $A \cdot B \cdot C \cdot D$

```
main.py x
1 def check_eligibility(A, B, C, D): 1 usage
2     # Eligible
3     eligible = A and B and C and (not D)
4
5     # Conditional
6     cond1 = (not A) and B and C and (not D)
7     cond2 = A and (not B) and C and (not D)
8     cond3 = A and B and (not C) and (not D)
9     cond4 = A and B and C and D
10
11     conditional = cond1 or cond2 or cond3 or cond4
12
13     # Not Eligible
14     not_eligible = not (eligible or conditional)
15
16     return eligible, conditional, not_eligible
17
18
19 # Test
20 def get_input(name): 4+ usages
21     value = input(f"{name} (1 = Yes, 0 = No): ")
22     return value == "1"
23
24 A = get_input("Attendance")
25 B = get_input("Assignments")
26 C = get_input("Fees")
27 D = get_input("Warning")
28
29 e, c, n = check_eligibility(A, B, C, D)
30
31 print("Eligible:", e)
32 print("Conditional:", c)
33 print("Not Eligible:", n)
```

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Example output shown
below

```
"C:\Users\oamma\PycharmProjects\Eligibility checker.py\.venv\Scripts\python.exe" "C:\Users\oamma\PycharmProjects\Eligibility checker.py\main.py"
Attendance (1 = Yes, 0 = No): 1
Assignments (1 = Yes, 0 = No): 1
Fees (1 = Yes, 0 = No): 1
Warning (1 = Yes, 0 = No): 0
Eligible: True
Conditional: False
Not Eligible: False

Process finished with exit code 0
```

“ AI / LLM Usage

We used ChatGPT to:

- help design Boolean logic
- verify truth table correctness
- generate test cases

All logic and implementation were understood and verified by the team.

CONCLUSION

Achievements:

- Successfully implemented digital logic + software
- Practical real-world application
- Good teamwork

Future Improvements:

- Add GUI interface
- More features (midterm marks, etc.)
- Real database connection

Demo Video

Screenshot of the CircuitVerse web simulator interface showing a digital logic circuit design.

Browser Tabs: Поиск | M365 Copilot, Group 10 Final 3.pptx, Омар Каримов - Circuits, CircuitVerse - Digital Circuits, Омар Каримов - Circuits, Final group 10 - Edit Project, EEE120-Group-10-Final-/, Iskandar.

Address Bar: circuitverse.org/simulator/edit/final-group-10-1f3d5891-95b2-48cf-a142-a93bf1d0a943

Toolbar: Project, Circuit, Tools, Help. Final group 10. Омар Каримов

Circuit Elements Panel: Search, Input, Output, Gates, Decoders & Plexers, Sequential Elements, Annotation, Misc.

TESTBENCH: A, Attendance, B, Assignments, C, Fees, D, Warning.

Timing Diagram: A, B, C, D, Attendance, Assignments, Fees, Warning.

Circuit Diagram: A logic circuit with multiple inputs (A, B, C, D, Attendance, Assignments, Fees, Warning) and outputs (Condition Eligible, Not Eligible, Eligible). The circuit uses various logic gates (AND, OR, NOT, XOR) and inverters.

Properties Panel: PROJECT PROPERTIES. Project: Final group 10. Circuit: Main. Clock Time (ms): 500. Clock Enabled: ☒. Lite Mode: ☐. Buttons: Edit Layout, Delete Circuit.

GitHub Link

GitHub Link